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INSTITUTE OF ENGINEERING MECHATRONICS DEPARTMENT(BE)

MINOR PROJECT PRESENTAION

Design, Fabrication and Performance Analysis of an Air Cushion Vehicle

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INTRODUCTION

- A hovercraft is an air cushion vehicle that moves on a layer of air below its body.
- The air layer reduces friction and allows smooth movement over land and water surfaces.
- The system uses one motor for lift generation and another motor for forward thrust.
- Direction control is achieved using a servo operated rudder mechanism.
- This project demonstrates the practical use of hovercraft technology.



Objective of Project

- The project was developed to understand the practical working of an RC controlled hovercraft system. The main objectives of the project are as follows:
- Design a compact and lightweight hovercraft body
- Generate lift using BLDC motor and propeller
- Provide forward motion using thrust motor
- Achieve steering through servo rudder
- Use RC transmitter and receiver for wireless control
- Study the performance of air cushion technology



COMPONENTS USED

- ❖ The hovercraft model was developed using electrical, mechanical, and control components. The main components used are listed below:

➤ Main Components:

- BLDC Motors
- ESC (Electronic Speed Controller)
- LiPo Battery
- Servo Motor
- RC Transmitter & RC Receiver
- Propellers
- XT60 Connector
- 3D Printed Body Parts

1. BLDC Motors (2)  • Generate lift force and forward thrust for movement of the hovercraft.	2. ESC (2)  • Controls and regulates the speed of both BLDC motors.	3. LiPo Battery  • Supplies electrical power to motors, receiver, ESC units, and servo system.	4. Servo Motor  • Controls rudder angle for steering and direction change.	5. RC Transmitter  • Sends user control commands wirelessly to the receiver.
6. RC Receiver  • Receives signals from transmitter and distributes them to ESC and servo.	7. Propellers  • Produce airflow required for lift generation and forward motion.	8. XT60 Connector  • Provides secure battery connection for safe power transfer.	9. PLA Filament  • PLA material used for 3D printing of hovercraft body parts.	10. 3D Printed Body Parts  • Supports mounting of all components and maintains body structure.

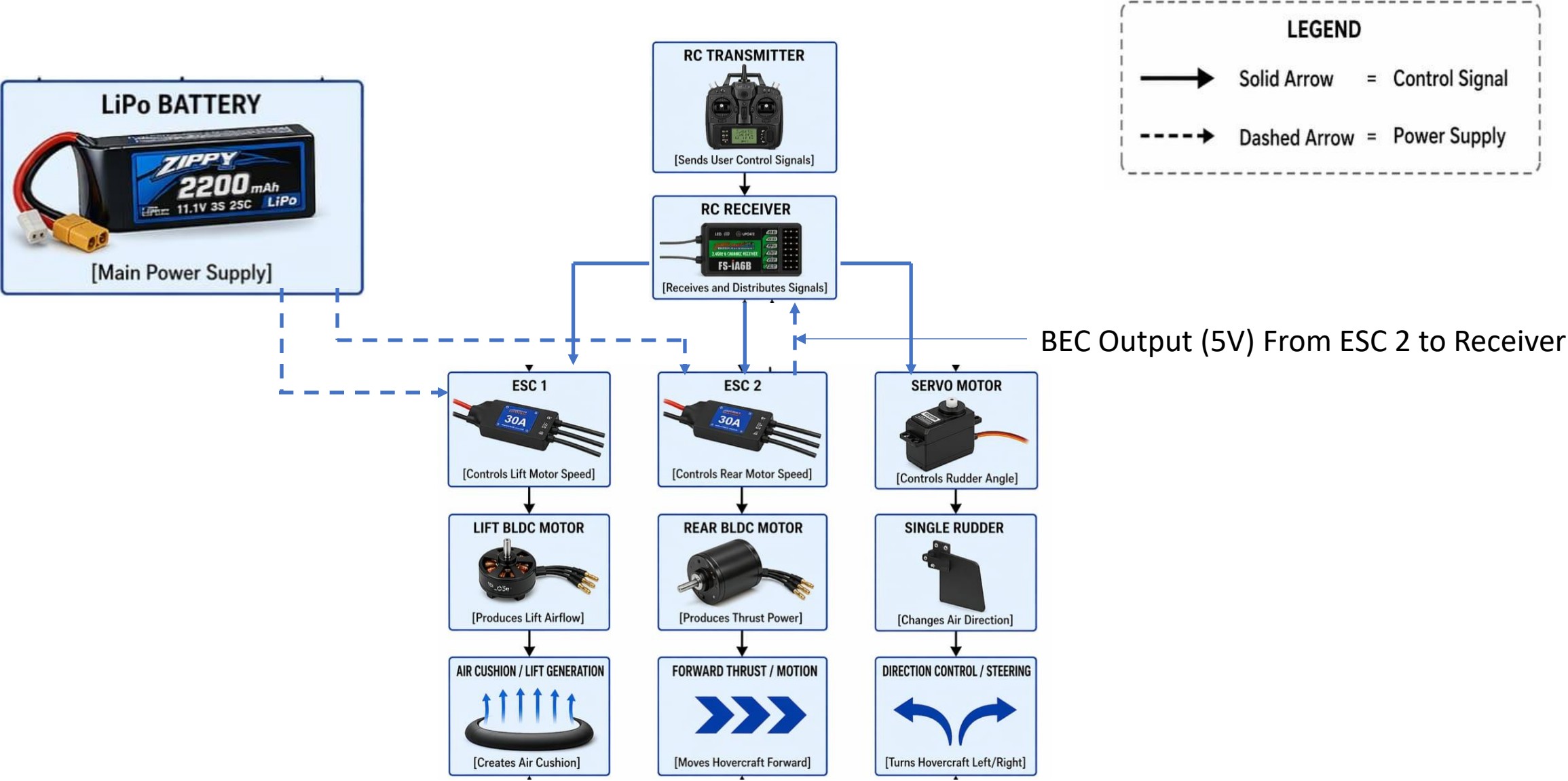
BLOCK DIAGRAM AND WORKING

❖ The working of the hovercraft is based on lift generation, forward thrust, and steering control. The system operation is explained below:

➤ Working Steps:

- LiPo battery supplies power to the ESC units and receiver system.
- RC transmitter sends control signals to the receiver wirelessly.
- Receiver distributes signals to both ESC units and servo motor.
- Lift motor rotates the propeller to create an air cushion below the body.
- Thrust motor generates forward force for movement of the hovercraft.
- Servo motor controls the rudder angle for left and right turning.
- Combined operation of lift, thrust, and steering enables smooth movement.

BLOCK DIAGRAM



Design and Fabrication Process

❖ The hovercraft body and supporting parts were designed and fabricated using suitable software tools and manufacturing methods.

➤ **The complete process is summarized below:**

- CAD modelling and body design using FreeCAD
- Slicing of final CAD model using Ultimaker Cura
- 3D printing of body parts using PLA material
- Assembly of motors, battery, ESC, and electronics
- Proper wiring and component alignment completed
- Final testing carried out after fabrication
- Smooth movement and system performance achieved

RESULTS AND CONCLUSION

❖ The hovercraft model was tested successfully under controlled conditions.

➤ **The important observations are given below:**

- Air cushion was generated successfully by lift motor
 - Forward movement achieved using thrust motor
 - Steering control responded properly through rudder system
 - Lightweight body improved movement efficiency
 - Wireless control system worked smoothly
 - Overall system performance was stable and satisfactory
- The project successfully demonstrated the design and working of an RC controlled air cushion vehicle.

THANK YOU

ANY QUESTION?